

Pipeline: Input to Verdict

SE498 Team 3 “Audio Rate” · from microphone and strap to the zone on the phone · state as of 2026-09-16

For future Claude

The data flow of the decided system, stage by stage, for the team: what enters each stage, what leaves it, which decision fixed it, and what can silently fail. Distilled from Audio-Derived Exercise Exertion Estimation §1–9 and the 2026-09-16 ADRs; nothing here is new design. The training path and the on-device path share the frozen front-end — that parity is the central engineering rule. Zone thresholds are [VERIFY] (ACSM-style bands, no opened source). Companion: Lab Session Runbook (how the inputs are produced), Data Management Plan (where they live).

The two paths

(diagram — open the note in Obsidian to see it rendered)

Offline (training) path: everything above, on a laptop, from session bundles. **On-device (verdict) path:** mic → qualification gate → frozen front-end → model → verdict, nothing uploaded. The strap, labels and lag fit exist only in the study.

Stage table (what renders in the PDF)

#	Stage	In	Out	Fixed by	Silent failure to watch
1	Capture	air near the mouth	16 kHz WAV per phone; IMU CSV; readbacks	design note §1; Decision - Handset roster three iPhones; Decision - IMU logged by the capture app	OS noise suppression or AGC re-enabled by an update — level is signal
2	Qualification gate	cal tone, room tone	pass/fail per phone per session	design note §2	a phone that passed last month fails today; checked every session
3	Reference stream	H10 over Polar SDK	RR intervals, raw ECG, chest ACC, timestamps	Decision - Breathing-rate reference H10 first belt conditional	dry electrodes at start; cadence-locked HR; only one device receives RR
4	Sync	clap in audio + RR artefact	per-phone offset to the strap clock	design note §3–4	BLE jitter and clock drift of seconds corrupt the lag fit itself

#	Stage	In	Out	Fixed by	Silent failure to watch
5	Labels	RR intervals, resting HR, age	%HRR at 1 Hz, smoothed ~10 s; Tanaka HRmax	Decision - Protocol one session per participant age-predicted HRmax	training on raw BPM → model learns identity; HRmax scatter ± 7–11 bpm is a constant per-person scale error
6	Lag fit	audio RMS envelope (300–3000 Hz, 1 s) × HR	one lag per participant, 0–90 s	Decision - Labels by fitted lag continuous exertion trajectory	lag outside the window or a flat correlation peak (Risk 10)
7	Windowing	aligned audio + %HRR	10 s windows, 5 s hop, label = mean %HRR; phase tag (ramp / steady / recovery); steady-state mask kept	same	windows straddling a stage transition without the phase tag
8	Front-end	window	64 log-mel frames, 25 ms / 10 ms, floor-clamped, normalised to the cal tone	design note §6	per-clip mean-variance normalisation (destroys level); app vs Python mel mismatch
9	Model	log-mel sequence, resting HR scalar	%HRR per window (+ breathing-rate auxiliary head)	design note §7	a per-window CNN without context underfits the lagged, smooth label
10	Evaluation	predictions vs labels	LOPO MAE; LODO MAE across three iPhone models ; MAE by ramp / steady / recovery; zone accuracy strict and ± 1; per-participant MAE; Bland–Altman; IMU-only and power-only baselines	Ground Truth and Validation; Proposal §4	LODO collapse = learned the microphone; LOPO collapse = learned the person
11	Verdict	window predictions	%HRR, zone, recent trend; EMA-smoothed; Core ML int8 on the phone	design note §9	quantised ≠float; app ≠Python on the same WAV — the file-injection parity test catches both
12	Incremental validity	Borg + differentiated ratings, HR, audio	within-subject mixed model, HR first, audio second	Decision - Target framing %HRR with incremental validity over HR	pooled correlations inflate; single-session RPE reliability is a stated limit (Risk 12)

From %HRR to a zone (verdict semantics)

Zone	%HRR band [VERIFY]	Shown as
very light	< 30	rest / warm-up

Zone	%HRR band [VERIFY]	Shown as
light	30–39	light
moderate	40–59	moderate — the rehab target band
vigorous	60–89	vigorous
near-maximal	≥ 90	back off

The app shows the current zone and a 60 s trend arrow, never a heart-rate number: the claims fence (Synthesis/Goal Framing Boundary) says zone-level guidance, not an HR monitor. Zone accuracy is scored strict and with a ± 1 -zone tolerance because the Tanaka denominator can shift a participant’s whole scale (Ground Truth and Validation).

Parity: the one rule that spans both paths

Same feature code, byte-identical parameters, in the training pipeline and the app. Verified three ways: numerical mel check against librosa on a fixed WAV; quantised-vs-float on test vectors; **file injection** — one WAV through the app and through Python must give the same prediction. Skipping this is the most common way live accuracy silently differs from offline accuracy (design note §6, §9).

Build order

Phase 0 instrument (capture app, qualification test, H10 ingestion incl. SDK ECG/ACC, IMU, feature-extractor port + parity) → Phase 1 pilot session one (belt check, lag-fit check) → Phase 1 cohort (< 10) → Phase 2 offline QA + labels → Phase 3 training + evaluation → Phase 4 on-device + live validation with a strap-wearing participant. Dates in Milestones.

Connections

- Audio-Derived Exercise Exertion Estimation · Audio Signal Processing and Features · Ground Truth and Validation · Lab Session Runbook · Data Management Plan · Milestones · Synthesis/Risk Register rows 8–12 · Results Ledger

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